

1. What is structural biology?

Answer:

Structural biology is the branch of molecular biology that focuses on understanding the **three-dimensional structure** of biological macromolecules (proteins, nucleic acids, etc.) and how these structures determine their **function**.

2. Why is structure important in biology?

Answer:

Structure determines how molecules interact with other molecules. The **function of a protein is intimately linked to its shape**, which influences binding, activity, and interactions with other biomolecules.

3. What are macromolecules?

Answer:

Macromolecules are large, complex molecules with high molecular weight, such as **proteins, nucleic acids (DNA/RNA), polysaccharides, and lipids**. Structural biology mainly focuses on proteins and nucleic acids.

4. What techniques are used in structural biology?

Answer:

Common techniques include:

- **X-ray crystallography**
- **Nuclear Magnetic Resonance (NMR)**
- **Cryo-Electron Microscopy (Cryo-EM)**
- **Small Angle X-ray Scattering (SAXS)**

5. What is the central dogma in the context of structural biology?

Answer:

The **central dogma** refers to the flow of genetic information:

DNA → RNA → Protein

Structural biology focuses on how this linear sequence (primary structure) folds into a **3D functional structure**.

6. What is meant by sequence-structure-function relationship?

Answer:

It refers to how the **amino acid sequence** (primary structure) determines the **3D structure** (secondary, tertiary, quaternary), which in turn governs the **biological function** of the molecule.

7. How are macromolecules represented in structural biology?

Answer:

Macromolecules are usually represented as **PDB (Protein Data Bank) files** containing atom coordinates in 3D space. Visualization tools like **PyMOL, Chimera, and Jmol** are used to view these structures.

8. What are the levels of protein structure?

Answer:

1. **Primary** – amino acid sequence
2. **Secondary** – local folding (α -helix, β -sheet)
3. **Tertiary** – 3D folding of entire chain
4. **Quaternary** – multiple chains (subunits)

9. What is the function of structural motifs and domains?

Answer:

Motifs are short, conserved sequences or structures that form recognizable shapes (e.g., helix-turn-helix). **Domains** are independently folded units within a protein, often with specific functions.

10. What does the term “active site” mean in proteins?

Answer:

The **active site** is the region of an enzyme where substrate binding and catalysis occur. It has a specific 3D structure crucial for enzymatic activity.

11. How can we read a protein structure file (PDB)?

Answer:

A PDB file includes **ATOM records** with x, y, z coordinates, residue names, atom names, chain IDs, and more. Specialized software is used to parse and visualize this data.

12. What is target-based drug design?

Answer:

It involves identifying a **biological target protein** (e.g., an enzyme in a disease pathway), determining its 3D structure, and designing small molecules to bind and inhibit/activate its function.

13. How do we identify a drug target protein?

Answer:

Targets are identified through:

- **Genomics/proteomics data**
- **Differential expression studies**
- **Pathway analysis**
- **Structural properties like active sites**

14. What is meant by protein structure prediction?

Answer:

It refers to computational methods (like **AlphaFold** or **homology modeling**) used to predict the 3D structure of a protein based on its amino acid sequence.

15. What is macromolecular crystallography used for?

Answer:

It determines the **atomic-resolution 3D structure** of a macromolecule by analyzing the diffraction pattern produced when X-rays pass through its crystal.

16. What is the role of Cryo-EM in structural biology?

Answer:

Cryo-Electron Microscopy allows visualization of **large and flexible complexes** in near-native states without needing crystals, especially useful for membrane proteins and viral particles.

17. How does NMR help in structural biology?

Answer:

NMR provides information about **atoms in a magnetic field** to deduce the **3D structure** of proteins in solution. It is especially useful for **small to medium-sized proteins**.

18. What is homology modeling?

Answer:

A method for predicting a protein's structure based on the **known structure of a homologous (similar) protein**, assuming conserved structure in evolution.

19. What is molecular docking?

Answer:

Docking simulates the interaction between a **protein and a ligand** (e.g., drug), predicting how they bind and the **binding affinity**.

20. What is functional annotation in protein structure studies?

Answer:

Functional annotation links a **protein structure** to its **biological role**, often by identifying active sites, motifs, domains, and structural homologs.